Cohort Mark Analysis (2026 vs 2025 vs 2024)

Thermodynamics and Combustion - Module

Methods / Data handling (for cohort comparison)

Cohort marks were analysed for three academic years (2026, 2025, 2024), stored as separate columns within a single matrix. Records with a value of **0** were treated as **missing / not-attempted** and excluded from all calculations and plots. Any values outside the valid range (**0–100%**) were also excluded to prevent invalid entries from influencing results.

Descriptive statistics are reported using both **mean ± standard deviation** (for comparability with conventional reporting) and **median with interquartile range (IQR)** (as a robust measure less sensitive to outliers). Percentiles are included to characterise performance across the full distribution (including lower and upper tails), rather than relying solely on a single central measure.

Visual comparisons combine **distribution plots** (histograms, box/swarm, density estimates, ECDFs) and **outcome-focused summaries** (grade band proportions and threshold attainment rates). Where a normal “bell curve” reference is displayed, it is included as an interpretive guide only; assessment marks are bounded and often structured by marking rubrics, and therefore should not be expected to follow a perfect normal distribution.

Grade band summaries use standard UK classification-aligned thresholds (Fail <40, Pass 40–49, 50–59, 60–69, ≥70) and are presented both as counts and as proportions to allow consistent year-to-year comparison irrespective of cohort size.

Summary statistics table (per cohort)

This table provides a compact numerical comparison of cohort performance. Mean and standard deviation indicate central tendency and spread, while median and interquartile range (IQR) provide robust equivalents that are less sensitive to outliers. Percentiles (P10, P25, P75, P90) highlight performance across the distribution (lower tail, centre, upper tail), supporting comparison of both typical performance and cohort variability across years.

```
=== Summary statistics (zeros ignored) ===
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Year	N	Mean	SD	Median	IQR	Min	P10	P25	P75	P90	Max	Skew
"2026"	59	51.542	18.594	57	24.75	11	18.8	40.25	65	73	76	-0.70114
"2025"	53	47.849	22.766	55	36.25	2	12.4	32	68.25	73	75	-0.59043
"2024"	36	48.333	17.398	46.5	23	3	24.1	41	64	71	76	-0.68519

Grade band counts and proportions

Grade-band summaries translate raw marks into outcome-relevant categories (e.g., fail, pass, 2:2, 2:1, first). Counts provide scale, while proportions allow direct year-to-year comparison even if cohort size differs. Changes in the distribution across grade bands are particularly informative for identifying whether year-to-year differences are primarily occurring around key thresholds (e.g., pass boundary) or at the upper end.

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=== Grade band counts ===
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Year	<40 Fail	40-49 Pass	50-59 (2:2)	60-69 (2:1)	70+ (1st)
"2026"	14	10	9	18	8
"2025"	14	11	7	11	10
"2024"	5	15	5	6	5

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=== Grade band proportions ===
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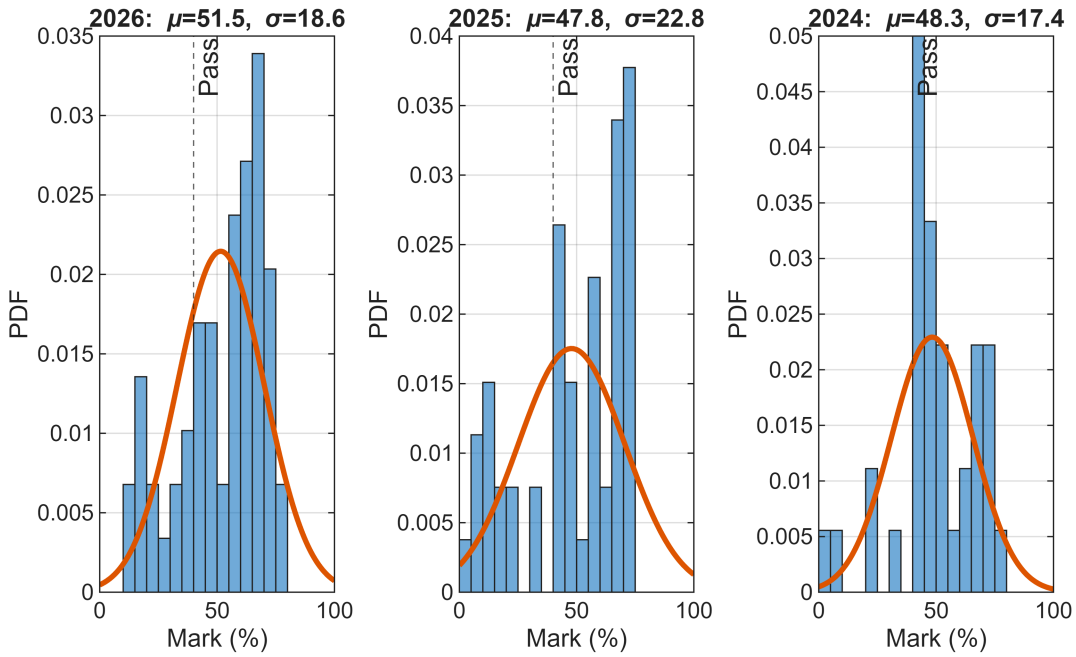
Year	<40 Fail	40-49 Pass	50-59 (2:2)	60-69 (2:1)	70+ (1st)
"2026"	0.23729	0.16949	0.15254	0.30508	0.13559
"2025"	0.26415	0.20755	0.13208	0.20755	0.18868
"2024"	0.13889	0.41667	0.13889	0.16667	0.13889

```
=== Pass rates ===
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Year	PassRate
"2026"	0.76271
"2025"	0.73585
"2024"	0.86111

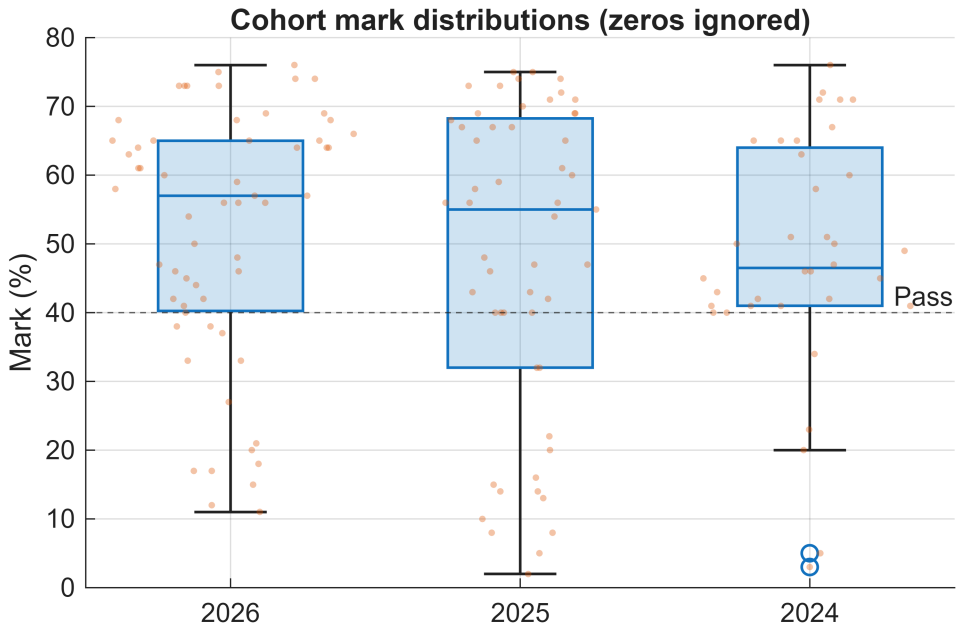
Histogram with normal reference (“bell curve” overlay)

This plot provides a direct view of the mark distribution by cohort. The overlaid normal curve uses the cohort’s mean and standard deviation as a reference, supporting a visual check of whether the overall shape is approximately symmetric and unimodal. Shifts in peak location indicate changes in typical performance, while differences in spread indicate changes in variability. Given marks are bounded (0–100) and often influenced by rubric structure, departures from the normal reference at extremes are expected and should not be interpreted as anomalous in isolation.



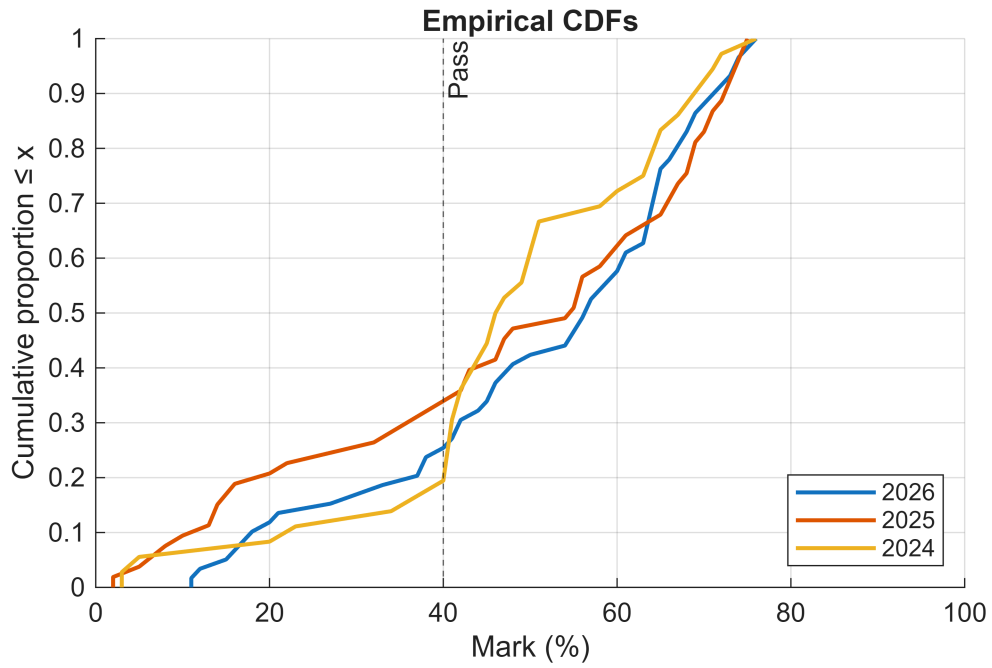
Boxplot with individual marks (swarm overlay)

This plot summarises cohort performance using medians and interquartile ranges while retaining visibility of individual marks. The median provides a robust measure of typical performance and the IQR indicates the central spread. The swarm overlay makes clustering and the presence of extreme outcomes more transparent than a summary-only plot, supporting moderation/consistency discussion (e.g., whether year-to-year changes are broad shifts across the cohort or driven by a small subset of marks).



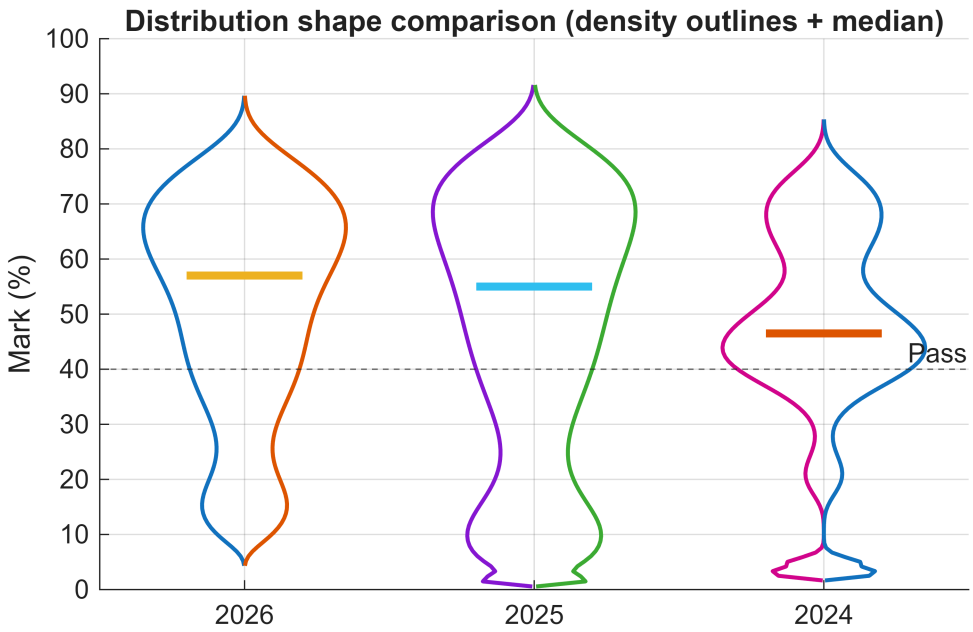
Empirical cumulative distribution function (ECDF)

The ECDF supports cohort comparison across the full mark range without relying on binning choices. For any mark value on the x-axis, the y-axis indicates the proportion of students scoring at or below that mark. Lower ECDF curves correspond to stronger overall performance (fewer students below each threshold). Reading at key thresholds (e.g., pass mark, 60%, 70%) provides an immediate cohort-level comparison of attainment across years.



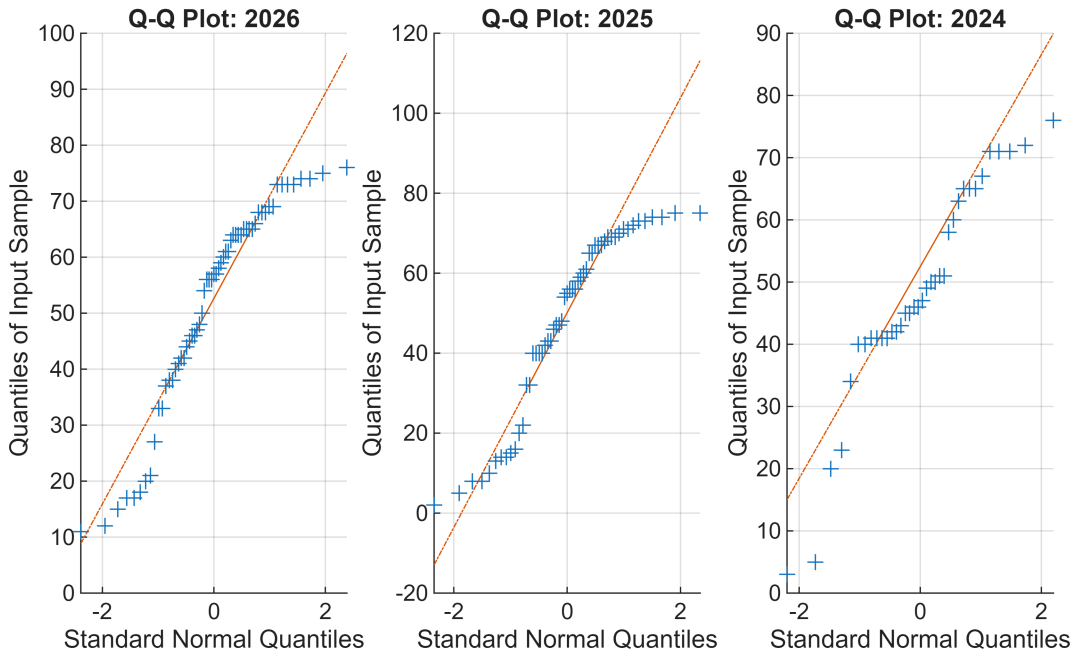
Kernel density comparison (“violin-style” outline with median)

This plot provides a smoothed view of distribution shape, allowing comparison of where marks are concentrated and whether there is evidence of clustering (e.g., around boundaries or common rubric outcomes). Wider regions correspond to higher density of marks in that range, and the median marker indicates typical performance. This is particularly useful for identifying multimodality or asymmetric shapes that may be less apparent in a boxplot.



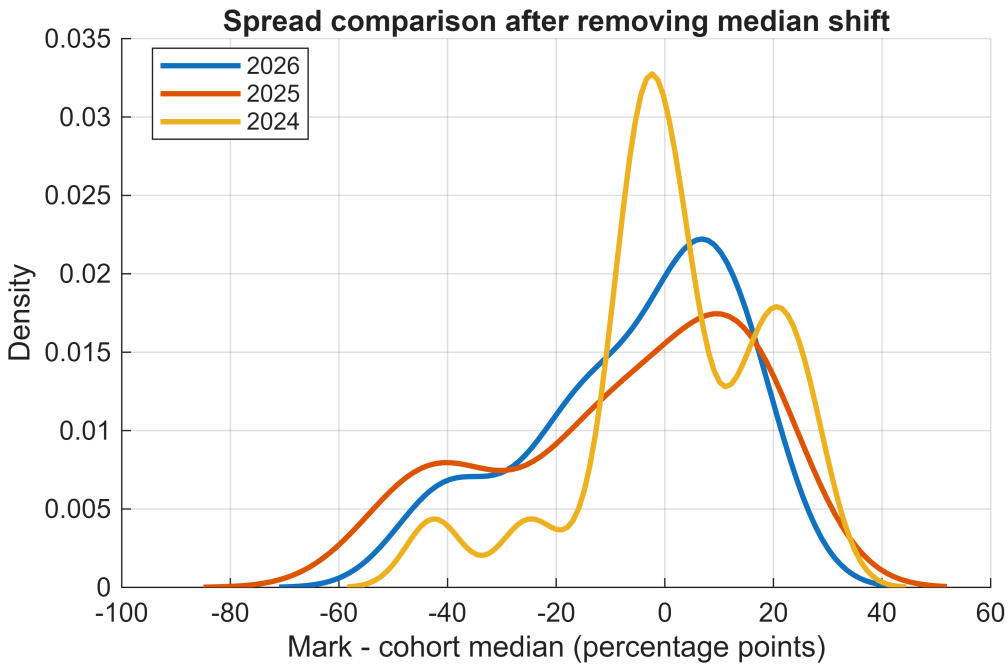
Q–Q plot (normality diagnostic)

This diagnostic assesses how closely the cohort distribution follows a normal distribution, particularly in the tails. Points close to the reference line indicate approximate normality in that region, while systematic departures indicate skewness, tail heaviness/lightness, or ceiling/floor effects. In assessment data, deviations at high/low marks and small “steps” (from discrete marking structure) are common; the plot is primarily included as a consistency/shape check rather than a requirement for subsequent comparisons.



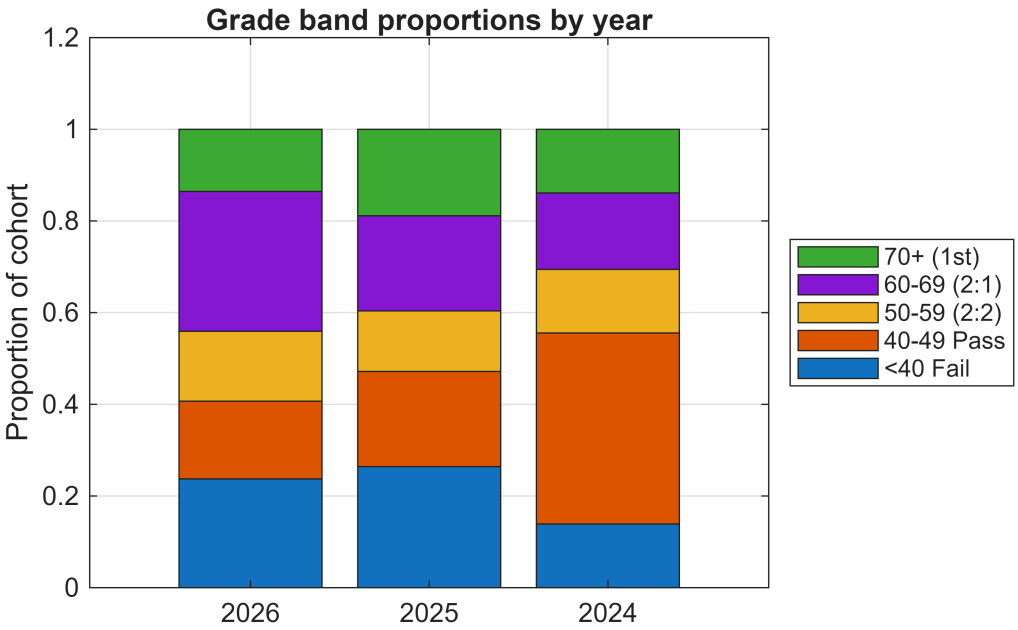
Recentered density comparison (mark minus cohort median)

This plot compares spread independently of differences in cohort median performance. Each cohort’s marks are shifted so that the median is at zero, then smoothed densities are overlaid. Similar recentered curves suggest cohorts differ mainly by an overall shift in performance, while differing widths indicate changes in variability. This supports interpretation of whether year-to-year differences reflect broad cohort movement or changes in the distribution’s dispersion.



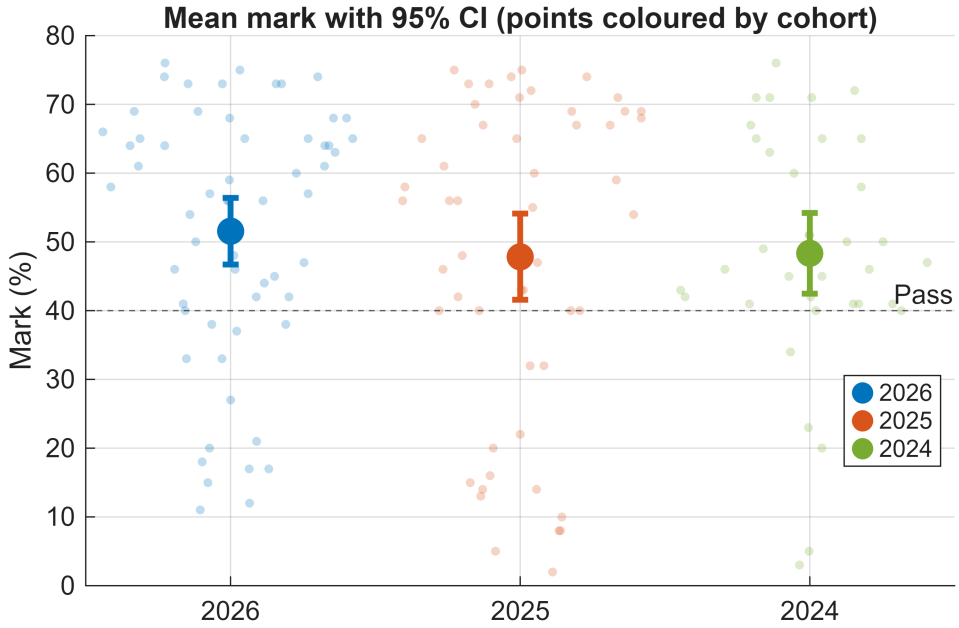
Grade band stacked proportions

This plot provides an outcome-oriented overview by showing the proportion of students in each grade band for each year. It supports rapid comparison of where cohort performance differs in a manner aligned with classification thresholds. It is especially useful for discussing changes at the pass boundary and shifts in the proportion of higher classifications (e.g., 2:1 and first-class outcomes) across cohorts.



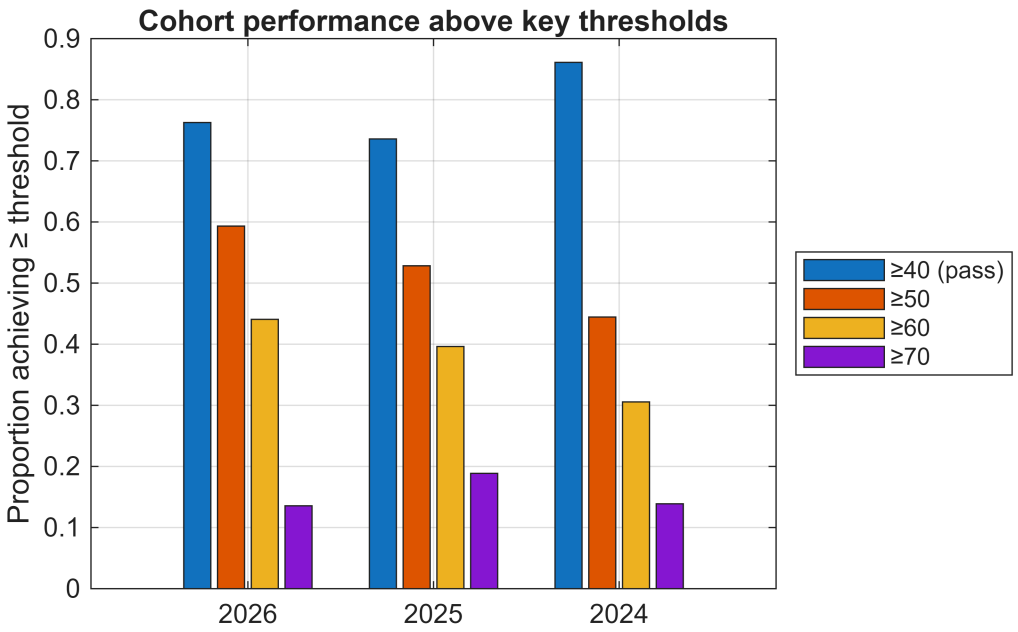
Mean with 95% confidence interval (with individual marks; coloured by cohort)

This figure summarises cohort averages with uncertainty intervals while retaining visibility of individual results. The confidence interval reflects uncertainty in the estimated cohort mean (accounting for sample size and dispersion), and should be interpreted as an interval estimate for the cohort mean rather than a spread of individual performance. Visual differences between cohorts should be considered alongside distributional plots and formal tests where relevant; overlap/non-overlap of confidence intervals provides a useful indication of effect magnitude but is not a substitute for statistical inference.



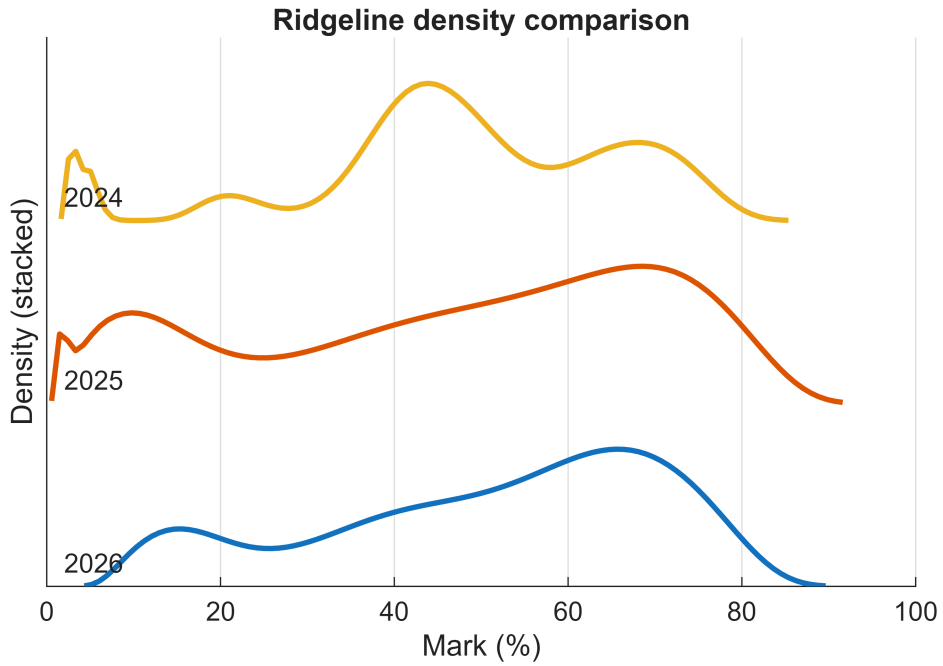
Proportion achieving key thresholds (≥40, ≥50, ≥60, ≥70)

This plot reports attainment rates at selected thresholds aligned to common academic decision points (pass, 2:2/2:1 boundary, first-class boundary). It provides a straightforward cohort comparison that is largely independent of distributional assumptions, highlighting whether changes are occurring mainly at the lower end (pass rate) or in progression into higher classifications.



Ridgeline density comparison

Ridgeline densities provide a presentation-friendly overview of distribution shape across years, allowing simultaneous comparison of typical mark regions (peak locations), spread, and tail behaviour. This is particularly suitable for reporting as it avoids overlap while retaining the intuitive “shape” comparison seen in density-based plots.




```
bullets = 5×1 string
"• Median (IQR): 2026 57.0 (24.8); 2025 55.0 (36.2); 2024 46.5 (23.0)."          ...
"• Mean ± SD: 2026 51.5 ± 18.6; 2025 47.8 ± 22.8; 2024 48.3 ± 17.4."
```

"• Pass rate (≥40%): 2026 76.3%; 2025 73.6%; 2024 86.1%."
"• Overall distribution is higher than 2024/25 (median +6.2 marks vs 2024/25 ave
"• Spread (IQR): 2026 is narrower than 2025 and wider than 2024."

TestTable = 3×5 table

	YearA	YearB	p_ranksum	p_Holm	Significant_0p05
1	"2026"	"2025"	0.5857	1	false
2	"2026"	"2024"	0.3278	0.9833	false
3	"2025"	"2024"	0.6973	0.6973	false

Kruskal–Wallis (overall): p = 0.6247. No strong evidence of a cohort difference (p ≥ 0.05).